SEVENTH ANNUAL ECC
UNDERGRADUATE
MATHEMATICS COMPETITION

April 3, 2004

Time control: Three hours

NO BOOKS, NOTES, CALCULATORS, COMPUTERS OR NON-TEAM-MEMBERS may be consulted.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper submitted.

Each problem counts 10 points. Partial credit for significant but incomplete work. Have fun!

1. Areas in arithmetic progression.

A circle of area A_1 is contained in a circle of area $A_1 + A_2$. If the radius of the larger circle is 3, and the three numbers $A_1,\ A_2,\ A_1 + A_2$ are in arithmetic progression, what is the radius of the smaller circle?

2. Roots of quadratics.

The roots of the equation $x^2 - ax + b = 0$ are r and s. If the roots of $x^2 - cx + d = 0$ are $r + \frac{1}{s}$ and $s + \frac{1}{r}$, find c and d in terms of a and b.

3. A radical equation.

Prove that if a, b, c, r, s and t are positive numbers such that

$$\frac{a}{r} = \frac{b}{s} = \frac{c}{t},$$

then

$$\sqrt{ar} + \sqrt{bs} + \sqrt{ct} = \sqrt{(a+b+c)(r+s+t)}.$$

4. A rational number.

Prove that

$$(\log_{64} 27)(\log_9 32)$$

is a rational number, and express it explicitly as a quotient of integers. (Reminder: no calculators. You may take the most basic properties of logarithms as given.)

5. Coin tossing.

A fair die is to be tossed repeatedly until a 6 occurs. If X is the number of tosses required, determine the smallest integer n for which

$$P(X \leq n) \geq \frac{1}{2}.$$

Justify your answer.

6. Which is larger?

Let f be a real-valued twice differentiable function on the positive reals satisfying

$$f''(x) > 0 \quad \text{for all } x > 0$$

. Let $h(x) = f(x) - xf'(x)$. Determine which is larger, $h(2003)$ or $h(2004)$.

7. A radical sum.

Evaluate $\lfloor S \rfloor$, where

$$S = \frac{1}{\sqrt{0}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{4}} + \cdots + \frac{1}{\sqrt{2002}+\sqrt{2004}},$$

and $\lfloor x \rfloor$, as usual, denotes the greatest integer not exceeding x. Justify your answer.

8. Limit of a sequence.

Evaluate

$$\lim_{n\to\infty} \sum_{k=1}^{n} \Big(1 + \frac{2k}{n}\Big) \ln\Big(1 + \frac{2k}{n}\Big)\Big(\frac{2}{n}\Big).$$

9. Conditions for a solution.

Assume that A, B, C and D are square matrices, all of the same size, and that $C + D$ and $C - D$ are nonsingular (i.e., invertible), but that A and B are singular (noninvertible). Show that necessary and sufficient conditions for the existence of square matrices X and Y satisfying the equations

$$AX + BY = C \quad \text{and} \quad BX + AY = D \tag{1}$$

are that $A + B$ and $A - B$ be nonsingular.

10. An upper bound.

Suppose that x, y and z are real numbers satisfying $x + y = z - 1$ and $xy = z^2 - 7z + 14$.
(a) Show that $x^2 + y^2 \leq 8$.
(b) Is $x^2 + y^2 = 8$ possible? Justify your answer.